

GED Mathematics Worksheet

Counting, Permutations, Combinations, and Probability

Instructions

Answer all 50 questions. Show your work where applicable. This worksheet covers fundamental concepts in counting methods and probability that appear on the GED Mathematics test.

1 Systematic Counting and Fundamental Counting Principle

1. A restaurant offers 4 different appetizers, 6 different main courses, and 3 different desserts. How many different three-course meals can be ordered (one item from each category)?
2. A license plate consists of 2 letters followed by 3 digits. How many different license plates can be made if repetition is allowed?
3. A school is assigning ID numbers that consist of 1 letter followed by 4 digits. How many different ID numbers are possible?
4. A clothing store stocks shirts in 5 colors, pants in 4 sizes, and shoes in 6 styles. How many different outfits can be created by selecting one shirt, one pair of pants, and one pair of shoes?
5. A password must consist of 2 uppercase letters followed by 3 digits. How many different passwords are possible if repetition is allowed?
6. At a sandwich shop, customers choose: bread (5 options), cheese (3 options), meat (4 options), and vegetables (6 options). How many different sandwiches can be made?
7. A survey has 5 yes/no questions. How many different ways can the survey be completed?
8. How many different outfits can be made with 3 hats, 4 shirts, and 5 pairs of pants?

9. A restaurant has 3 types of soda, 2 types of juice, and 4 types of water. How many different beverage choices are available?
10. A code consists of one color (from 8 available), followed by one number (from 0-9), followed by one letter (from 26 available). How many different codes are possible?

2 Permutations

11. In how many ways can 5 students be arranged in a line?
12. A coach is choosing a captain, co-captain, and treasurer from a team of 8 players. How many ways can these three positions be filled?
13. How many different 4-letter arrangements can be made from the letters A, B, C, and D (using each letter exactly once)?
14. In how many ways can 6 people be seated in 6 chairs around a conference table?
15. A competition has 10 participants. In how many ways can first, second, and third place be awarded?
16. A playlist has 7 different songs. How many different ways can the songs be arranged?
17. How many different 3-digit codes can be made from the digits 1, 2, 3, 4, 5, and 6 (without repetition)?
18. A teacher is arranging 4 students in a row for a photo. How many different arrangements are possible?
19. A company needs to select a president, vice-president, and secretary from 9 employees. In how many ways can this be done?
20. How many 5-letter arrangements can be made from the letters in the word WORLD (using each letter at most once)?

3 Combinations

21. A committee of 3 people is being chosen from a group of 8 people. How many different committees can be formed?
22. How many ways can 4 books be selected from a shelf of 10 books?
23. In a lottery, 6 numbers are chosen from 1 to 49. How many different lottery tickets are possible?
24. A student must choose 2 elective classes from 5 available electives. How many different combinations of electives are possible?

25. How many different 5-card hands can be dealt from a standard 52-card deck?
26. A team is choosing 4 starters from a roster of 12 players. How many ways can the starting team be selected?
27. How many ways can 3 toppings be selected from 8 available pizza toppings?
28. A store is selecting 6 products from a catalog of 20 products to feature in an advertisement. How many different selections are possible?
29. How many ways can a person choose 2 vegetables from a list of 7 vegetables?
30. A committee needs to select 5 members from a group of 15 volunteers. How many different committees are possible?

4 Distinguishing Between Permutations and Combinations

31. Is the following a permutation or combination problem? “How many ways can 3 students be chosen to represent the class?”
32. Is the following a permutation or combination problem? “In how many ways can 4 people be arranged in a line?”
33. A student is choosing 5 books from 8 books to take on a trip. Is this a permutation or combination? How many ways can this be done?
34. A principal is selecting a class president, vice-president, and secretary from 20 students. Is this a permutation or combination? How many ways can this be done?
35. In how many ways can a coach select 6 players from 14 for a game? Is this a permutation or combination?
36. Is it a permutation or combination to arrange 7 distinct paintings on a wall?

5 Basic Probability

37. A fair die is rolled. What is the probability of rolling a 4?
38. A bag contains 5 red marbles and 3 blue marbles. If one marble is drawn at random, what is the probability that it is red?
39. A spinner is divided into 8 equal sections numbered 1 through 8. What is the probability of spinning a number less than 5?
40. A standard deck of cards has 52 cards. What is the probability of drawing a heart?
41. A coin is flipped. What is the probability of getting tails?

42. A box contains 4 green, 6 yellow, and 5 red balls. If one ball is randomly selected, what is the probability that it is yellow?
43. A number is randomly selected from 1 to 20. What is the probability that it is a multiple of 3?
44. A fair die is rolled. What is the probability of rolling a number greater than 4?
45. A bag contains 8 candies: 3 chocolate and 5 vanilla. What is the probability of randomly selecting a vanilla candy?
46. A spinner has 10 equal sections: 4 are blue, 3 are red, and 3 are green. What is the probability of spinning blue?

6 Independent Events and Dependent Events

47. A fair coin is flipped twice. What is the probability of getting heads both times? (These are independent events.)
48. A bag contains 4 red marbles and 6 blue marbles. If two marbles are drawn with replacement, what is the probability that both are red?
49. Two cards are drawn from a standard deck without replacement. What is the probability that both are hearts?
50. A die is rolled, and then a coin is flipped. What is the probability of rolling a 3 and flipping heads?
51. A bag contains 3 green and 5 yellow balls. If two balls are drawn without replacement, what is the probability that both are yellow?